

Software Partnership Proposal: Chamber of Gold Buyers Ghana

1. Industry Context (why this is a good time to move)

Ghana's gold trading sector is mid-formalization:

- **GoldBod (Ghana Gold Board)**, established under Act 1140 (2025), is now the sole authorized buyer, assayer, seller, and exporter of gold from licensed ASM (artisanal/small-scale mining) producers.
- Licensed buyers operate on a **Tier 1 / Tier 2** structure — Tier 1 buys directly from miners and finances them; Tier 2 buys from Tier 1.
- Industry bodies (Chamber of Licensed Gold Buyers, Chamber of Bullion Traders Ghana) are **publicly and repeatedly calling for**: digital traceability systems, ISO/LBMA-compliant assay labs, better access to finance for miners, and downstream value-addition (refining, jewelry).
- ASM production jumped 70% in 2024 and is a growing share of national output — meaning transaction volume, compliance burden, and data complexity are all increasing fast.

This means the Chamber's members (and the Chamber itself as a body) have an active, stated need for exactly the kind of software your company builds. The pitch isn't "here's a nice-to-have app" — it's "here's the infrastructure the industry has already said it needs."

2. Partnership Models to Propose

A. Chamber-as-anchor-client (association-level contract)

The Chamber licenses a shared platform (traceability, member registry, pricing feed) for all its members, funded via membership dues or a subscription tier. You become the Chamber's official technology partner/vendor of record — strong credibility for future contracts with GoldBod, Minerals Commission, or individual buyers.

B. Individual buyer/member contracts (B2B SaaS)

Once you have the Chamber's endorsement, license per-buyer software (inventory, compliance, POS-style buying desk tools) directly to Tier 1/Tier 2 licensed buyers as a recurring SaaS product. The Chamber's endorsement becomes your distribution channel.

C. Regulator-facing partnership (GoldBod / Minerals Commission)

Position a traceability/reporting layer that plugs into GoldBod's reporting requirements. If you can become the "compliance rail" buyers use to report to GoldBod, you become close to indispensable — GoldBod has explicitly said it wants better traceability tooling.

D. Financial services integration

Partner with banks, mobile money providers (MTN MoMo, Telecel Cash), or microfinance institutions to build the financing/payment layer miners and buyers need (the Chamber has explicitly called for "access to finance" for ASM miners). You build the rails; a bank or fintech provides capital and licensing.

E. Assay lab / logistics tech partnership

Chamber has called for ISO/LBMA-compliant assay labs. Partner with an assay lab operator (or the Chamber itself) to build the lab information management + certification software, positioning your firm at a technical layer regulators trust.

F. Data/analytics partnership with international buyers or exporters

Once traceability data exists, it has resale/reporting value to international refiners and LBMA-aligned buyers who need provenance data for compliance. This is a longer-term revenue stream but worth structuring the data model to support it from day one.

3. Detailed Software Systems to Build

System 1: Gold Trade Traceability & Chain-of-Custody Platform

The flagship product — this is the single most-requested capability in the sector right now.

- **Core function:** Track a unit of gold from miner → Tier 1 buyer → Tier 2 buyer → GoldBod/export, with every transaction, weight, purity assay, and price recorded immutably.
- **Key features:**
 - Digital "gold passport" per lot (QR/barcode or RFID-tagged sealed bags), recording origin site, miner ID, weight, purity, date, GPS coordinates of purchase point.
 - Chain-of-custody ledger — each transfer of ownership logged with timestamp, both parties' license numbers, and price paid.
 - Offline-first mobile app for buying agents in remote/low-connectivity mining areas, syncing when back online.
 - Tamper-evidence: cryptographic hashing of each record (a private/permissioned blockchain or simply an append-only signed ledger — doesn't need to be a public blockchain to get the integrity benefits).
 - Automated compliance reports formatted for GoldBod submission.
 - Smuggling-risk flags: anomaly detection when volumes/prices deviate from regional norms.
- **Tech stack suggestion:** React Native or Flutter for offline mobile capture; PostgreSQL with append-only audit tables (or Hyperledger Fabric if a permissioned ledger is

wanted); Node.js/Django backend; SMS fallback (via Africa's Talking or similar) for ultra-low-connectivity areas.

- **Why it wins:** Directly answers the Chamber's own public ask. Whoever builds this becomes systemically important.

System 2: Buyer Operations & Compliance Suite ("Gold Desk")

A day-to-day operating system for licensed Tier 1/Tier 2 buying houses.

- **Core function:** Digitize the buying desk — replace paper ledgers and manual scales logs.
- **Key features:**
 - Live weighing-scale integration (Bluetooth/USB scale input) with tamper-proof weight capture.
 - Live gold price feed integration (world market price + GoldBod's published buying rate).
 - Automatic calculation of miner payout at the correct % of world price (GoldBod has pushed transparent pricing — up to 98% of world price to miners).
 - Miner/supplier registry with license verification lookup against GoldBod/Minerals Commission databases.
 - Cash/mobile money payout reconciliation.
 - Daily/weekly/monthly automated regulatory filings.
 - Multi-branch support for buyers with several buying stations across regions (Ashanti, Western, Central, Northern).
- **Tech stack:** Web dashboard (React) + tablet-based POS app; integrates with local mobile money APIs; role-based access for buying agents vs. branch managers vs. compliance officers.

System 3: Miner Finance & Credit Platform

Answers the Chamber's stated call for "access to finance and technology" for ASM miners.

- **Core function:** Let licensed Tier 1 buyers (or partner banks) extend working-capital advances/pre-financing to miners against future gold delivery, digitally tracked.
- **Key features:**
 - Digital miner profiles with transaction history (feeds credit scoring — miners with a longer clean trading history unlock better advance terms).
 - Loan/advance origination, repayment-via-gold-delivery tracking.
 - Integration with mobile money for disbursement.
 - Risk dashboard for buyers/lenders showing exposure per miner/region.
- **Partnership angle:** Build this jointly with a bank or fintech that provides the actual capital; you own the software and take a platform fee or revenue share.

System 4: Assay Lab Information Management System (LIMS)

Supports the Chamber's push for ISO/LBMA-compliant assay labs.

- **Core function:** Manage sample intake, testing workflows, and certificate issuance for gold purity assays.
- **Key features:**
 - Sample chain-of-custody from intake to result.
 - Integration with assay equipment (XRF analyzers, fire assay results entry).
 - Digital, verifiable certificates (QR-code verifiable) tied back to the trade traceability platform (System 1).
 - Audit trail for ISO/LBMA compliance reviews.
- **Why it matters:** Assay results are the single most disputed data point in gold trading — a trusted digital certificate system builds credibility for the whole Chamber's membership internationally.

System 5: Chamber Membership & Governance Portal

A lower-cost, high-goodwill starting project — good as a first deliverable to build trust before larger contracts.

- **Core function:** Member registry, license status tracking, dues/payments, meeting/AGM management, member communications, and a public-facing directory of verified licensed buyers (useful for combating impersonation/fraud by unlicensed buyers).
- **Key features:**
 - Member self-service portal (renew license documentation, pay dues, download compliance certificates).
 - Public verification lookup ("is this buyer licensed?") — valuable for miners and the public to avoid scams.
 - Automated alerts to members about regulatory changes (EPA fees, GoldBod policy updates, licensing deadlines).
 - Event/training management for capacity-building programs.

System 6: Market Price & Analytics Dashboard

- **Core function:** Aggregate and display live gold prices (world market, GoldBod buying rate, regional average buying prices) plus historical trend data for members.
- **Key features:**
 - Live price ticker (integrate a gold price API + GoldBod's published rates).
 - Regional price comparison to flag underpayment/exploitation of miners.
 - Export volume and FX inflow dashboards (useful for the Chamber's advocacy work with GoldBod/Bank of Ghana — they already cite these figures publicly).

- Anti-smuggling analytics: identify regions with production/export volume mismatches.

System 7: Downstream Value-Addition Marketplace

Supports the Chamber's stated interest in local refining, fabrication, and jewelry production.

- **Core function:** A B2B marketplace connecting licensed buyers/refiners with local jewelers and fabricators, plus inventory/order management for that supply chain.
- **Key features:**
 - Verified-gold sourcing marketplace (buyers list refined gold lots with traceability certificates attached).
 - Order and inventory management for jewelry manufacturers.
 - Export documentation assistant for jewelry/fabricated gold exporters.

4. Suggested Build & Contract Sequencing

Phase	System	Rationale
1 (Trust-building)	Membership & Governance Portal (System 5)	Low complexity, fast delivery, builds relationship and credibility
2 (Core value)	Buyer Operations Suite (System 2) + Market Price Dashboard (System 6)	Immediate day-to-day value for individual member buyers, becomes revenue-generating SaaS
3 (Flagship/strategic)	Traceability & Chain-of-Custody Platform (System 1)	Positions you as critical infrastructure; strongest pitch to GoldBod/regulators for wider adoption
4 (Expansion)	Assay LIMS (System 4), Miner Finance Platform (System 3)	Requires partner relationships (assay labs, banks) — build once trust and data foundation exist
5 (Long-term)	Value-Addition Marketplace (System 7)	Monetizes the traceability data and network effects built in earlier phases

5. Commercial Structure Ideas

- **Chamber contract:** annual platform license fee (membership portal + market dashboard bundled) paid by the Chamber centrally.

- **Per-buyer SaaS:** monthly subscription per buying station for the Operations Suite, tiered by transaction volume.
 - **Transaction/traceability fee:** small per-lot fee on the traceability platform, potentially shared with GoldBod if adopted as official infrastructure.
 - **Financing platform:** revenue share or platform fee with partner bank/fintech, not direct lending risk on your side.
 - **Data/reporting products:** premium analytics subscription for international buyers/refiners wanting provenance data (long-term).
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Suggested next step

A short discovery workshop with Chamber leadership to validate which pain point is most urgent right now (traceability compliance vs. day-to-day buyer operations vs. member services) — that should decide which system becomes Phase 1.